

SANTA MARIA, BrazilWMO: **839370**Lat: **29.711S**Lon: **53.688W**Elev: **88**StdP: **100.28**Time Zone: **-3.00 (SAB)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.2	4.0	-0.1	3.8	6.2	1.5	4.2	6.6	10.3	24.2	8.9	23.7	0.7	120	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	36.0	24.3	34.2	24.0	33.0	23.9	26.8	31.8	26.1	31.1	25.4	30.2	3.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	21.1	29.7	24.9	20.1	28.9	24.1	19.1	28.2	85.0	32.3	81.5	31.1	78.4	30.2	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.3	6.3	-0.8	38.9	1.2	0.8	-1.7	39.4	-2.4	39.9	-3.0	40.4	-3.9	41.0		
			DB													
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	25.6	25.5	23.3	20.5	17.1	14.7	14.0	16.3	17.5	20.0	22.4	24.7
	DBStd	5.52	2.73	2.89	3.12	3.62	3.85	4.71	4.98	5.09	4.13	3.31	3.07	2.87
	HDD10.0	31	0	0	0	0	2	9	15	5	1	0	0	0
	HDD18.3	568	0	0	3	19	68	132	156	103	62	19	4	1
	CDD10.0	3719	484	434	412	315	223	149	137	200	227	311	373	454
	CDD18.3	1216	227	201	156	85	31	22	20	40	39	71	128	197
	CDH23.3	10401	2080	1772	1194	594	194	144	152	396	335	526	1143	1871
	CDH26.7	4101	904	731	448	203	48	27	24	127	131	183	439	837
Wind	WSAvg	2.6	2.8	2.6	2.4	2.0	2.1	2.3	2.5	2.9	3.0	3.1	3.0	2.8
Precipitation	PrecAvg	1790	163	150	128	149	128	131	153	108	171	202	142	170
	PrecMax	2730	441	294	300	343	288	263	264	263	348	431	442	491
	PrecMin	1113	64	46	14	34	8	5	52	20	56	55	26	1
	PrecStd	359	101	69	73	72	66	59	70	53	79	94	97	114
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	37.2	36.9	34.0	30.9	29.1	29.0	32.1	34.1	35.2	36.2	37.4
		MCWB	24.6	25.0	24.5	22.1	21.9	20.9	19.3	21.0	21.2	22.1	21.9	23.9
	2%	DB	35.2	34.8	33.5	31.1	28.0	27.1	27.1	30.0	30.2	31.8	33.2	35.0
		MCWB	24.9	26.0	24.7	22.3	20.4	19.9	18.8	20.3	20.3	22.3	22.3	23.3
	5%	DB	33.7	32.8	31.2	29.1	25.5	24.9	25.0	27.8	27.0	28.1	31.1	33.0
		MCWB	25.1	25.4	24.2	21.8	19.9	19.1	18.1	19.4	19.9	20.8	21.3	23.2
	10%	DB	31.8	30.9	29.1	26.9	23.0	22.1	22.0	25.2	24.1	26.0	29.1	31.2
		MCWB	24.4	24.8	23.2	21.1	19.1	18.3	17.5	18.6	18.8	20.0	20.8	22.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	27.9	27.8	24.9	23.5	21.9	20.9	22.5	23.4	24.6	25.1	26.7
		MCDB	34.0	32.2	31.8	30.1	27.3	26.9	25.7	29.2	29.4	31.7	30.9	32.0
	2%	WB	26.7	26.9	26.1	23.7	21.8	20.7	19.7	21.0	21.7	23.2	23.7	25.8
		MCDB	32.1	31.5	30.7	28.6	25.4	25.4	25.1	27.7	27.4	29.4	29.2	31.5
	5%	WB	25.9	26.1	25.1	22.7	20.7	19.6	18.7	20.0	20.7	22.0	22.9	24.9
		MCDB	31.1	30.6	29.3	26.8	24.0	23.8	23.7	26.1	25.5	26.5	28.2	30.3
	10%	WB	25.1	25.4	24.1	21.8	19.7	18.5	17.6	19.0	19.6	21.0	22.1	24.0
		MCDB	29.7	29.6	27.9	25.4	22.4	22.1	21.5	24.4	23.6	24.8	27.2	29.3
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.0	10.6	11.0	9.5	9.3	10.0	11.0	10.1	10.6	11.9	11.6
		MCDBR	13.3	11.9	13.1	13.7	12.6	11.5	13.0	13.9	14.7	14.0	15.2	14.5
		MCWBR	5.1	5.2	5.8	6.1	6.5	6.1	6.9	6.4	7.1	6.5	6.3	5.8
	5% WB	MCDBR	11.0	10.5	11.4	11.1	10.2	9.9	11.5	12.8	12.1	12.1	12.1	12.2
		MCWBR	5.1	5.2	6.0	5.4	5.9	5.6	6.6	6.8	6.5	6.3	5.9	5.6
Clear-Sky Solar Irradiance	taub		0.422	0.413	0.399	0.393	0.365	0.388	0.390	0.490	0.519	0.460	0.404	0.424
	taud		2.342	2.372	2.400	2.382	2.433	2.360	2.326	1.992	1.937	2.158	2.338	2.309
	Ebn at Noon		921	911	890	841	817	764	780	729	763	857	931	922
	Edh at Noon		135	127	116	107	92	94	101	156	182	155	134	140
All-Sky Solar Radiation	RadAvg		6.57	6.05	5.14	3.99	2.93	2.44	2.71	3.42	4.14	5.11	6.41	6.76
	RadStd		0.55	0.38	0.40	0.39	0.31	0.21	0.31	0.30	0.42	0.65	0.76	0.62

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page